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Regional disparities in incidence and outcomes of invasive ductal carcinoma of the breast among Asian and Pacific Islander women in the United States

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ABSTRACT

Background: Asian women have experienced a disproportionate increase in breast cancer incidence over the past four decades when compared to patients of other races. We aim to determine the variation of incidence and survival rates for Asian women based on their residential status by region of the United States.

Methods: The Surveillance, Epidemiology and End Results (SEER) 17-State database was used to identify cases of ductal carcinoma and its subtypes (ICD-O-3/3 codes: 8500, 8521, 8503, 8507, 8514, 8522, 8523; C50.0–50.9) among Asian and Pacific Islander (API) women during 2000–2020. Chi square tests were used for comparison of clinical and socioeconomic variables and Kolmogorov-Smirnov and Kruskal Wallis tests were used to assess differences between mean time to treatment and diagnosis. Incidence was analyzed via Joinpoint Regression Software with Kaplan-Meier survival curves for evaluation of survival by region, measured in months. Multivariate Cox proportional hazards regression identified independent predictors of survival. All statistical analyses were conducted using SPSS Version 29.0.2, with significance at $p < 0.05$.

Results: The West represented the largest incidence over the 20-year study period with 89.6 per 100,000 (95 % CI: 88.9–90.2). There were notable differences in both 5-year and 10-year survival rates with the south having the largest decline, dropping from 84 % at 5 years and 54 % at 10 years. This is despite the south being found to have the shortest time to treatment, 0.97 months (95 % CI: 0.95–0.99; $p = 0.009$). Multivariate Cox regression showed API women in the South had a 24 % increased risk of mortality compared to those in the West (aHR 1.244, 95 % CI: 1.176–1.315; $p < 0.001$).

Conclusions: The results of this study demonstrate significant disparities in incidence, treatment patterns, and survival outcomes amongst API women by residential status.

1. Background

Breast cancer incidence in the United States has increased by an average of 0.5 % annually over the past four decades, partly due to improved detection through screening programs [1,2]. The rate of increase, however, varies significantly among different racial groups, with Asian populations experiencing a notable rise in breast cancer incidence despite traditionally low rates [3,4]. This disparity may be partly explained by lower breast cancer screening rates among Asian Americans compared to other racial groups, even when accounting for education, insurance, and income [5]. These lower screening rates can lead to delayed diagnoses and, consequently, poorer survival outcomes [6].

Despite significant advances in breast cancer research, Asian women remain under-represented in large scale studies [7], contributing to disparities. These disparities are compounded by unique challenges, including the diverse genetic origins encompassed within the “Asian” racial classification. This heterogeneity, along with geographic barriers and reduced access to care in underserved regions, contributes to delayed diagnosis and limited screening, often resulting in the use of more aggressive treatments and lower survival rates. Geographic location is a known determinant of reduced access to timely diagnosis and treatment, further exacerbating these inequalities [8]. Collectively,

Abbreviations: API, Asian Pacific Islander; SEER, The Surveillance, Epidemiology and End Results.

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these factors underscore the need for targeted efforts to improve health equity for this population.

This study aims to identify these gaps by using data from the Surveillance, Epidemiology, and End Results (SEER) database to explore how geographic location influences incidence, treatment, and survival outcomes of invasive ductal carcinoma (IDC) among Asian and Pacific Islander women in the United States. By examining regional variations and their interplay with patient demographics, tumor characteristics, and access to care, this work seeks to provide a deeper understanding of the factors contributing to inequities in breast cancer outcomes. The findings aim to inform targeted interventions to improve survival and reduce disparities within this under-represented population.

2. Methods

This retrospective cohort study utilized the Surveillance, Epidemiology, and End Results [9] 17-State Research Plus Registry Database to identify cases Fof ductal carcinoma among Asian and Pacific Islander (API) women from 2000 to 2020. Patients from the SEER Research Plus Database who were identified as API, with pathologically confirmed ductal carcinoma and its subtypes as defined by ICD-O-3 histology and site codes (8500, 8521, 8503, 8507, 8514, 8522, 8523; C50.0–50.9) located in the breast (primary site codes: C50.0–50.9) were included. Data from individuals in California, Georgia, Connecticut, Seattle (Puget Sound), Utah, Louisiana, Kentucky, Hawaii, New Mexico, Iowa, and New Jersey were available for this study and grouped by region in the United States, modeling widely agreed upon geographic divisions such as those used by National Geographic [10]. The West included Hawaii, Seattle (Puget Sound), Utah, Alaska Natives, San Jose-Monterey (SJM), Los Angeles, and California excluding San Francisco/SJM, and Los Angeles, while the South included Georgia, Louisiana, and Kentucky. The Midwest was represented by Iowa, the east of Connecticut and New Jersey, and the Southwest alongside New Mexico. Demographics of interest included age, marital status, median household income (adjusted to 2021 dollars), and rural/urban residence. Tumor features included histologic subtype, SEER combined stage, breast-adjusted American Joint Committee on Cancer (AJCC) stage, and positivity of estrogen, progesterone, and HER2 receptors. Rates of metastasis to lung, bone, brain, and liver were obtained to assess spread between regions. Treatment was assessed by type of surgery and rates of chemotherapy and radiation. Chi-square tests were used to compare the clinical and socioeconomic variables among United States regions.

Age-adjusted incidence rates were calculated using SEER*Stat software, while Joinpoint Regression software utilized annual percent changes to identify directional shifts in incidence by region. Kruskal-Wallis testing was used to assess differences in mean time from diagnosis to treatment (in months). Univariate Kaplan-Meier survival curves were generated to evaluate differences in survival by region. Multivariate Cox regression was used to identify location as an independent predictor of survival when adjusting for age, income, histologic subtype, stage, and ER, PR, and HER2 statuses. All statistical analyses were completed using SPSS (Version 29.0.2), with significance set at $p < 0.05$.

3. Results

A total of 83,913 patients in the Surveillance, Epidemiology, and End Results Database were identified as meeting the inclusion criteria for analysis. The majority were from the West, which accounted for 50,763 patients (60.5 %), followed by the South with 13,525 patients (16.1 %). The East contributed 6146 patients (7.3 %), while the Southwest and Midwest had significantly smaller representations, with 172 patients (0.2 %) and 279 patients (0.3 %), respectively. Over the 20-year study period, the overall incidence of invasive ductal carcinoma was 87.1 cases per 100,000 people (95 % CI: 86.5–87.7). Regional variations were evident, with the West reporting the highest incidence at 89.6 per 100,000 (95 % CI: 88.9–90.2) [Table 1]. The East followed with an incidence of 78.8 per 100,000 (95 % CI: 76.9–80.6), while the Southwest reported 70.2 per 100,000 (95 % CI: 65.3–75.4) [Table 1]. The Midwest and South had the lowest incidences, at 76.9per 100,000 (95 % CI: 60.2–76.3) and 66.6 per 100,000 (95 % CI: 64.2–69.1), respectively [Table 1]. These findings highlight significant regional differences in IDC incidence across the United States.

The Kaplan-Meier survival analysis revealed significant regional differences in survival ($p < 0.001$) [Fig. 1]. The West demonstrated a 5-year survival rate of 87 %, decreasing to 64 % at 10 years, while the East had a slightly higher 5-year survival rate of 88 %, declining to 69 % at 10 years. The South exhibited the largest decline, with survival rates dropping from 84 % at 5 years to 54 % at 10 years, a difference of 30 percentage points [Fig. 1]. The Midwest and Southwest had similar survival rates, with 5-year rates of 85 % and 83 %, respectively, both declining to 64 % at 10 years [Fig. 1]. Multivariate Cox regression showed that API women in the South had a 24 % increased risk of mortality compared to those in the West (aHR 1.216, 95 % CI: 1.160–1.275; $p < 0.001$) 1), independent of clinical and socioeconomic factors [Fig. 2].

There were some differences in clinicopathological features. Intraductal papillary adenocarcinoma was rarely seen universally but was highest in the West (0.5 %). Similarly, micropapillary carcinoma was rarely seen but was highest in the Midwest (1.6 %) [Table 2]. Slightly larger differences were seen amongst the infiltrating duct and lobular carcinoma or infiltrating duct mixed with other types of carcinoma

Table 1
Demographic Characteristics of Invasive Ductal Carcinoma Among API Women by Region in the United States.

Demographic Characteristics	East	West	South	Southwest	Midwest	P-value
Overall Incidence						
Incidence per 100,000 (95 % CI)	78.8 (76.9–80.6)	89.6 (88.9–90.2)	66.6 (64.2–69.1)	70.2 (65.3–75.4)	76.9 (60.2–76.3)	
Age (years)						
< 40	653 (9.0 %)	4504 (7.5 %)	955 (5.9 %)	14 (6.4 %)	41 (12.8 %)	**< .001**
40–49	2033 (28.1 %)	13862 (23.1 %)	2875 (17.8 %)	50 (22.7 %)	84 (26.2 %)	
50–59	1953 (27.0 %)	16184 (27.0 %)	3991 (24.7 %)	66 (30.0 %)	86 (26.8 %)	
60–69	1532 (21.2 %)	14273 (23.8 %)	4147 (25.6 %)	53 (24.1 %)	67 (20.9 %)	
70–79	841 (11.6 %)	7895 (13.2 %)	2766 (17.1 %)	27 (12.3 %)	33 (10.3 %)	
> 80	211 (2.9 %)	3245 (5.4 %)	1452 (9.0 %)	10 (4.5 %)	10 (3.1 %)	
Married at diagnosis						
Yes	5116 (74.9 %)	39167 (68.2 %)	9616 (61.6 %)	126 (65.6 %)	228 (72.4 %)	**< 0.001**
No	1713 (25.1 %)	18256 (31.8 %)	5999 (38.4 %)	66 (34.4 %)	87 (27.6 %)	
Median Household Income (inflation adjusted to 2021)						
< 75 K	1429 (19.8 %)	23673 (39.5 %)	3665 (22.6 %)	216 (98.2 %)	297 (92.5 %)	**< 0.001**
> 75 K	5793 (80.2 %)	36290 (39.5 %)	12520 (77.4 %)	4 (1.8 %)	24 (7.5 %)	
Rural/Urban Living Status						
Rural	26 (0.4 %)	266 (0.4 %)	2354 (14.5 %)	43 (19.5 %)	76 (23.7 %)	**< 0.001**
Urban	7196 (99.6 %)	59697 (99.6 %)	13831 (85.5 %)	177 (80.5 %)	245 (76.3 %)	

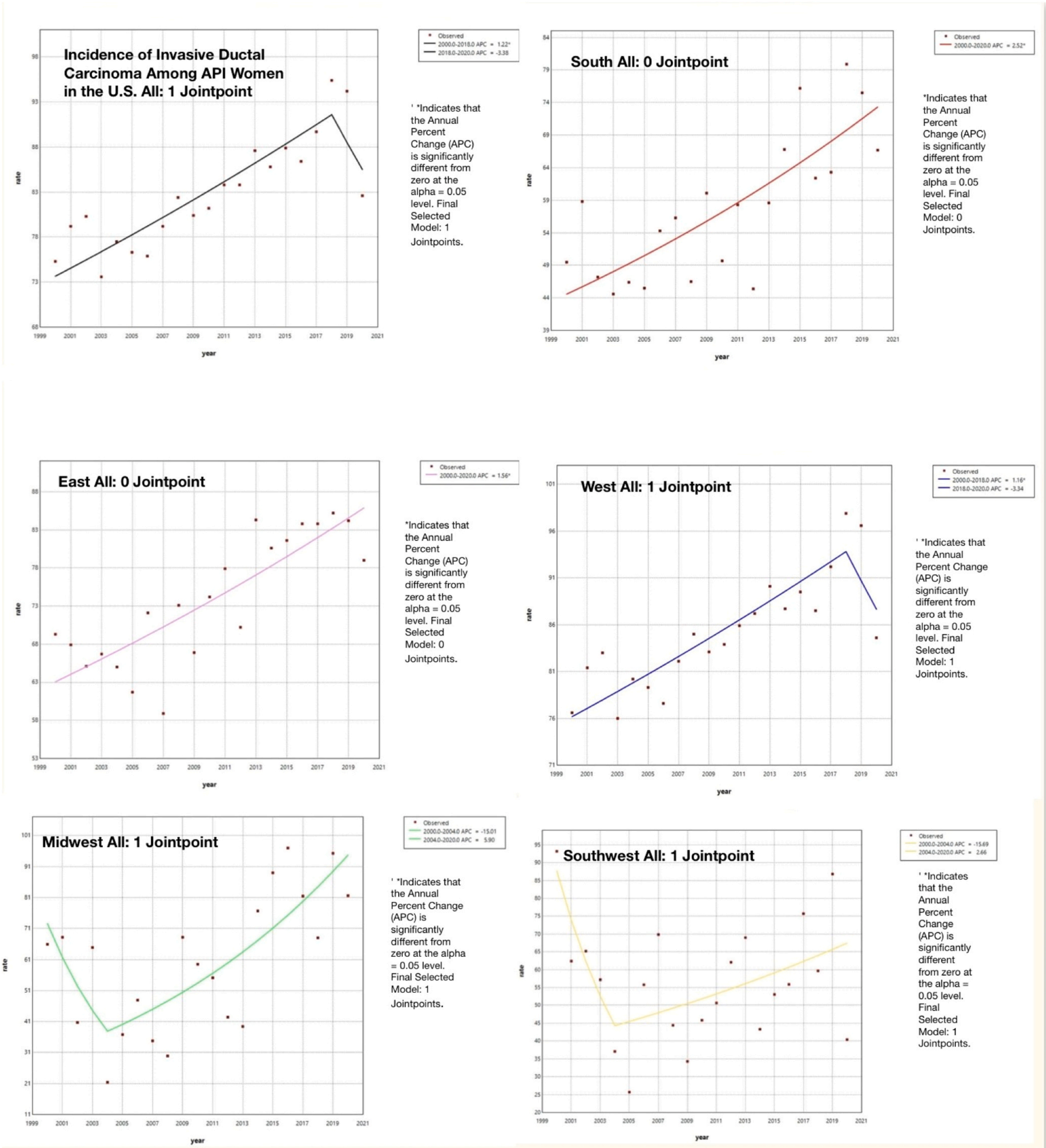

Fig. 1. “Incidence of Ductal Carcinoma of the Breast Among Asian Women in Various Regions of the United States from 2000 to 2020”.

subtypes [Table 2]. The outer quadrant of the breast or unspecified/overlapping locations were the two most common locations for all regions ($p < 0.001$) [Table 2]. ER, PR, and HER2 status were all similar across regions with most patients having ER and PR positivity and only the minority having HER2 ($p < 0.001$) [Table 2].

Living in the South was associated with a higher stage at diagnosis ($p = 0.03$), while patients in the Midwest had the highest rates of cancer-directed surgical intervention ($p = 0.02$) [Table 2]. Patients in all regions were more likely to present without metastasis to the lungs, brain, liver, or bones at initial diagnosis [Table 3]. Amongst all anatomical locations listed, the Midwest had the highest percentage of patients without metastasis ($p < 0.001$) [Table 3]. All regions were noted to have a higher number of patients who were married at diagnosis than not, as well as more patients living in an urban setting than rural ($p < 0.001$) [Table 1]. There was greater variation when looking at median household income, with the East and South having higher rates of malignancy in households making more than $75,000, the Southwest and Midwest having greater rates in households making less than $75,000, and equal rates for the west ($p < 0.001$) [Table 1].

The average time from diagnosis to treatment varied significantly

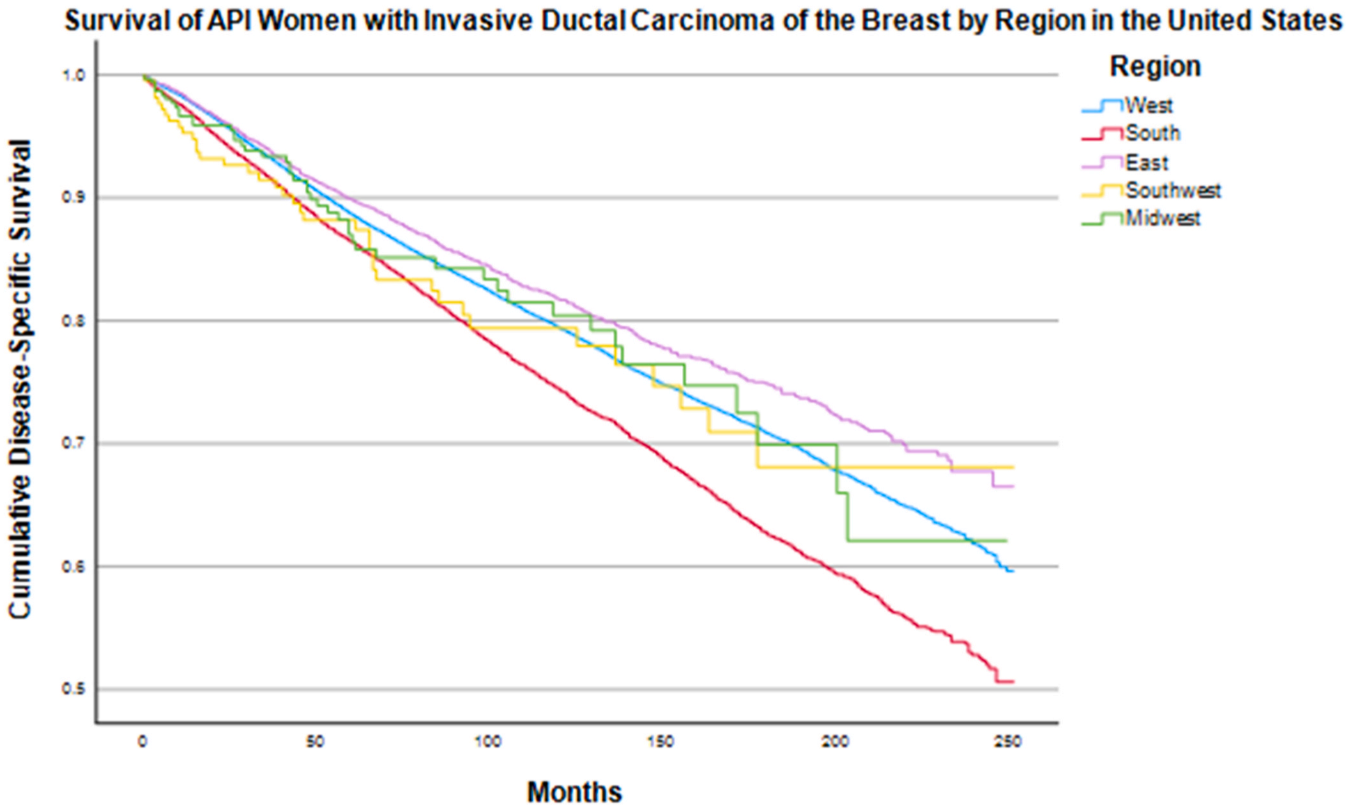

Fig. 2. "Kaplan-Meier survival curve of Asian women with ductal carcinoma of the breast by US geographic location".

Table 2
Cancer Characteristics of Invasive Ductal Carcinoma Among API Women by Region in the United States.

Cancer Characteristics	East	West	South	Southwest	Midwest	P-value
Histologic Subtype						
Intraductal papillary adenocarcinoma	31 (0.4 %)	277 (0.5 %)	41 (0.3 %)	0 (0.0 %)	1 (0.3 %)	**< 0.001**
Micropapillary ductal carcinoma	16 (0.2 %)	273 (0.5 %)	47 (0.3 %)	2 (0.9 %)	5 (1.6 %)	
Desmoplastic type ductal carcinoma	1 (0.0 %)	0 (0.0 %)	0 (0.0 %)	0 (0.0 %)	0 (0.0 %)	
Infiltrating duct and lobular carcinoma	498 (6.9 %)	3639 (6.1 %)	508 (3.1 %0	14 (6.4 %)	7 (2.2 %)	
Infiltrating duct mixed with other types of carcinoma	276 (3.8 %)	1789 (3.0 %)	277 (1.7 %)	3 (1.4 %)	8 (2.5 %)	
Primary Site						
Central portion of the breast	415 (5.7 %)	3294 (5.5 %)	1122 (6.9 %)	10 (4.5 %)	21 (6.5 %)	**< 0.001**
Inner quadrants of the breast	1291 (17.9 %)	11306 (18.9 %)	3200 (19.8 %)	47 (21.4 %)	54 (16.8 %)	
Outer quadrants of the breast	2741 (37.9 %)	23326 (38.9 %)	6581 (40.7 %)	92 (41.8 %)	128 (39.9 %)	
Unspecified/overlapping location	2776 (38.4 %)	22037 (36.8 %)	5283 (32.6 %)	71 (32.3 %)	118 (36.8 %)	
Central portion of the breast	415 (5.7 %)	3294 (5.5 %)	1122 (6.9 %)	10 (4.5 %)	21 (6.5 %)	
SEER Stage						
Distant	296 (4.7 %)	2220 (4.2 %)	614 (4.5 %)	11 (6.0 %)	11 (3.9 %)	**< 0.001**
Regional	1955 (30.8 %)	16060 (30.6 %)	3692 (26.9 %)	48 (26.2 %)	92 (32.3 %)	
Localized	4087 (64.5 %)	34265 (65.2 %)	9433 (68.7 %)	48 (26.2 %)	92 (32.3 %)	
Breast- Adjusted AJCC 6th Stage						
0	0 (0.0 %)	0 (0.0 %)	0 (0.0 %)	0 (0.0 %)	0 (0.0 %)	**< 0.001**
1	2039 (46.7 %)	18092 (47.3 %)	5557 (52.1 %)	65 (50.0 %)	75 (43.4 %)	
2a, 2b	1557 (35.6 %)	14465 (37.8 %)	3630 (34.0 %)	43 (33.1 %)	64 (37.0 %)	
3a, 3b, 3c	598 (13.7 %)	4416 (11.5 %)	1093 (10.2 %)	13 (10 %)	30 (17.3 %)	
4	176 (4.0 %)	1307 (3.4 %)	389 (3.6 %)	9 (6.9 %)	4 (2.3 %)	
Receptor Positivity						
ER	5374 (79.4 %)	46803 (81.5 %)	13112 (83.0 %)	165 (81.3 %)	263 (84.3 %)	**< 0.001**
PR	4687 (69.6 %)	40216 (70.8 %)	11557 (73.5 %)	145 (72.1 %)	239 (76.6 %)	**< 0.001**
HER2	946 (13.1 %)	7593 (12.7 %)	1597 (9.9 %)	24 (10.9 %)	43 (13.4 %)	**< 0.001**

ER- estrogen receptor, PR- progesterone receptor, HER2- human epidermal growth factor receptor 2

depending on region. Patients in the Midwest had the shortest time to treatment, with an average of 0.85 months (95 % CI: 0.72–0.97) [Table 5]. The South and Southwest were tied with an average of 0.97 months (95 % CI: 0.95–0.99 and 0.84–1.09, respectively). The East reported an average of 1.11 months (95 % CI: 1.09–1.14), and the longest wait times for treatment after initial diagnosis were found in the West (1.12 months; 95 % CI: 1.11–1.13) [Table 5]. The South was the least likely to not undergo an operation for treatment of their cancer (6.0 %) and the most likely to have a partial mastectomy (54.6 %) compared to its counterparts [Table 4]. The East and West had nearly identical rates of radiation at 92.9 % and 92.8 % respectively [Table 4].

Table 3
Presence of Metastasis at Diagnosis of Invasive Ductal Carcinoma Among API Women by Region in the United States.

Presence of Metastasis	East	West	South	Southwest	Midwest	P-value
Presence of lung metastasis						
Yes	79 (1.1 %)	560 (0.9 %)	157 (1.0 %)	0 (0.0 %)	3 (0.9 %)	**< .001**
No	4786 (66.3 %)	38221 (63.7 %)	9643 (59.6 %)	139 (63.2 %)	233 (72.6 %)	
Presence of brain metastasis						
Yes	13 (0.2 %)	93 (0.2 %)	40 (0.2 %)	1 (0.5 %)	1 (0.3 %)	**< .001**
No	4852 (67.2 %)	38689 (64.5 %)	9756 (60.3 %)	138 (62.7 %)	235 (73.2 %)	
Presence of liver metastasis						
Yes	54 (0.7 %)	453 (0.8 %)	113 (0.7 %)	5 (2.3 %)	3 (0.9 %)	**< .001**
No	4813 (66.6 %)	38342 (63.9 %)	9689 (59.9 %)	134 (60.9 %)	233 (72.6 %)	
Presence of bone metastasis						
Yes	126 (1.7 %)	1007 (1.7 %)	275 (1.7 %)	4 (1.8 %)	3 (0.9 %)	**< .001**
No	4739 (65.6 %)	37804 (63.0 %)	9528 (58.9 %)	135 (61.4 %)	233 (72.6 %)	

Table 4
Type of Treatment for Invasive Ductal Carcinoma Among API Women by Region in the United States.

Type of Treatment	East	West	South	Southwest	Midwest	P-value
Surgery						
No Surgery	646 (9.0 %)	4087 (6.8 %)	955 (6.0 %)	29 (13.2 %)	24 (7.5 %)	**< .001**
Partial mastectomy	3433 (47.7 %)	29041 (48.4 %)	8728 (54.6 %)	109 (49.8 %)	171 (53.3 %)	
Total mastectomy	1949 (27.1 %)	15120 (25.2 %)	3421 (21.4 %)	43 (19.6 %)	52 (16.2 %)	
Radical mastectomy	1115 (15.5 %)	11238 (18.7 %)	2820 (17.6 %)	35 (16.0 %)	70 (21.8 %)	
Bilateral mastectomy	2 (0.0 %)	10 (0.0 %)	1 (0.0 %)	0 (0.0 %)	0 (0.0 %)	
Mastectomy, NOS	59 (0.8 %)	446 (0.8 %)	55 (0.4 %)	3 (1.4 %)	4 (1.2 %)	
Radiation						
Received Radiation therapy	3266 (92.9 %)	27762 (91.9 %)	9002 (92.8 %)	106 (96.4 %)	178 (93.2 %)	**< .001**
Chemotherapy						
No	3817 (52.8 %)	33149 (55.3 %)	9685 (59.8 %)	121 (55.0 %)	164 (51.1 %)	**< .001**
Yes	3406 (47.2 %)	26814 (44.7 %)	6501 (40.2 %)	99 (45.0 %)	157 (48.9 %)	

NOS-Not otherwise specified

Table 5
Variation in time from diagnosis to treatment of Invasive Ductal Carcinoma Among API Women by Region in the United States.

	East	West	South	Southwest	Midwest
Time from diagnosis to treatment (months)					
Mean (95 % CI)	1.11 (1.09–1.14)	1.12 (1.11–1.13)	0.97 (0.95–0.99)	0.97 (0.84–1.09)	0.85 (0.72–0.97)
Median [IQR]	1.00 [0.00–2.00]	1.00 [0.00–2.00]	1.00 [0.00, 1.00]	1.00 [0.00–1.00]	1.00 [0.00–1.00]

IQR: Interquartile range

4. Discussion

The highest incidence of IDC was observed in the Western region of the United States which may be attributable to multiple factors. Prior studies have shown that increased screening via mammography is directly related to higher incidence reports [11]. The Western region has greater access to mammograms and an aggressive state-level screening prevalence when compared to other U.S. regions [12]. Additionally, mammography rates tend to be greater in urban areas, and the West has a greater proportion of urban populations when compared to the South or Midwest [13]. While accessible screening is the most established cause for higher incidence, less studied possibilities could be environmental risk factors or lifestyle factors more prevalent in this area.

Despite shorter duration of time to treatment in the South, survival outcomes were notably worse, suggesting that factors beyond treatment timing, such as stage at diagnosis or the quality of treatment received, may play a critical role in long term outcomes. It is important to consider that the confidence intervals for time to treatment did overlap, however it is still worth noting as this measure provides a useful descriptive overview of how care is being delivered across regions. Through the data presented in this paper, the South was seen to have more advanced disease at diagnosis, consistent with lower screening rates. Although cancer-directed surgical intervention was more frequently employed in the South, survival rates declined, visible by the greatest survival drop from 5 years to 10 years relative to the other regions. A discrepancy in treatment modality, distance to providers, or limited specialists could also contribute.

Socioeconomic and geographic factors likely further contribute to the observed gaps. The Southern population has a higher representation of rural and low-income individuals, two groups that are associated with limited access to healthcare, delayed diagnosis, and high rates of comorbidities [14]. Further, the South has a dense population of racial and ethnic minorities who have historically faced disparities and inequities throughout the healthcare system. Together, these factors could aid in explaining the region's poorer survival outcomes despite a shorter initial time to treatment.

In contrast, patients in the East exhibited better survival outcomes than those in the West, even with longer times to treatment. This suggests that other factors, such as healthcare quality, multidisciplinary care approaches, or patient support systems, may play a compensatory role. Higher median household incomes and greater access to comprehensive cancer care in the East could also contribute to these improved outcomes. These observations underscore the importance of not just timely treatment but also effective and coordinated care in improving survival rates.

These results are consistent with existing hypotheses regarding racial and regional disparities in breast cancer outcomes, which attribute differences to variations in tumor biology, comorbidities, and systemic inequities.

5. Limitations

This paper was limited by a few factors. The nature of the study is retrospective, making it vulnerable to over generalization and inability to demonstrate a cause-effect relationship. This database does not represent the entirety of the United States, currently only accounting for nearly 48 % of the total population [15]. Although this study focuses on geographic location, SEER does not currently account for individuals who have recently changed location, making opportunity for false conclusions. Additionally, the data that has been used for this study utilized a limited number of locations which resulted in some regions having a higher density of participants compared to others. Two regions, Midwest and Southwest, are represented by one state due to registry availability. The SEER database does not give insight into factors that could potentially influence treatment decisions or choices regarding screening, which could incorrectly estimate gaps in care.

6. Conclusion

This study found significant regional disparities in the incidence, treatment patterns, and survival outcomes of invasive ductal carcinoma (IDC) among Asian and Pacific Islander (API) women in the United States. While the overall incidence of IDC has increased over the 20-year study period, survival outcomes varied notably across regions, underscoring the complex interplay between geographic, socioeconomic, and clinical factors.

By focusing specifically on API women, this study provides valuable insights into an underrepresented population, adding to the understanding of how disparities manifest within this diverse group. Prior studies have often focused on White and Black populations, making this an important step toward addressing gaps in the literature.

In conclusion, this study underscores the need for region-specific interventions to address disparities in IDC outcomes among API women. Expanding access to early detection, addressing systemic barriers to care, and improving treatment quality are critical priorities, particularly in regions with the poorest outcomes. Further research should explore the biological, cultural, and systemic factors influencing these disparities to inform targeted strategies. By prioritizing equity in cancer care, we can work toward improving outcomes for API women diagnosed with IDC.

CRediT authorship contribution statement

Gowri Warikoo: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis. **Bianca Ituarte:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Morales-Ramirez Pedro:** Visualization, Validation, Supervision, Project administration, Formal analysis. **Kailee Bunte:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Patient Consent

Not applicable

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article. The authors declare the following financial or non-financial interests which may be considered as potential conflicts of interest.

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